

Reflective LLM Prompt Optimisation for Interpreting GNN Predictions in Traffic Forecasting

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Abstract—Graph Neural Networks (GNNs) are effective for traffic forecasting, but their predictions are difficult to interpret. While post-hoc explanation tools like GNNExplainer generate useful importance scores across graph nodes and edges, the raw outputs are hard to interpret without additional context. Large Language Models (LLMs) offer a way to translate these scores into natural language, but often fluent yet unfaithful explanations—even when the input scores are invalid. We propose a two-phase, multi-agent reflective framework to address these limitations. Phase 1 optimises a prompt that enables LLMs to assess the validity of GNN importance scores using synthetic labelled data. Phase 2 incorporates this optimised prompt and focuses on refining the explanation component of the prompt to produce more faithful narratives grounded in multimodal traffic context. Experiments show that our framework improves the LLM’s ability to validate input scores before generating explanations as well as the ability to generate more trustworthy, context-aware explanations grounded in multimodal evidence. The framework provides a structured path toward more adaptable and dependable LLM-based interpretation of complex machine learning models in critical traffic prediction applications.

Index Terms—Explainable AI (XAI), Graph Neural Networks (GNNs), Large Language Models (LLMs), Multi-Agent Systems (MAS), Prompt Engineering, Reflection, Trustworthy AI, Intelligent Transportation Systems (ITS), Traffic Analysis.

I. INTRODUCTION

Graph Neural Networks (GNNs) are increasingly used in intelligent transportation systems (ITS) for tasks like traffic prediction due to their ability to model complex road relationships [1]–[3]. However, their ‘black box’ nature limits trust and deployment in safety-critical applications requiring interpretable model reasoning [4], [5]. Explainable AI (XAI) techniques specifically designed for GNN models have been introduced to better explain GNN predictions. For instance, *GNNExplainer* [6] produces importance scores for edges and node features, highlighting which parts of the graph structure and attributes are most influential in a GNN’s prediction [7]. However, these raw scores are often abstract and difficult to interpret without additional context, especially for traffic

professionals unfamiliar with GNN internals. Understanding these explanations requires combining insights from network data, textual descriptions, and visual maps [8], which has traditionally depended on human experts. However, relying on human experts to translate scores into actionable insights is time consuming and does not scale effectively [9].

Large Language Models (LLMs) offer a promising alternative to human experts for interpreting machine learning outputs by translating numerical values into coherent natural language explanations based on their contextual reasoning strengths [10], [11]. A key limitation in using LLMs as such a translator or an interface between models and human operators is their tendency to generate fluent and convincing narratives even when the input data is flawed. For instance, LLMs may attempt to rationalise any given GNN importance scores without verifying whether those scores actually align with the real-world contexts. This poses a serious risk in practical settings, where users may inadvertently supply incorrect inputs. Therefore, the core challenge is not just translating GNN importance scores into natural language, but ensuring that the explanations are grounded in valid reasoning. This shifts the focus towards designing and optimising the prompt to guide the LLM in both assessing the plausibility of the scores and, only when appropriate, generating faithful explanations.

To address this challenge, this paper proposes a novel **multi-agent reflective framework** that performs systematic, automated prompt refinement via iterative self-reflection in a collaborative agent system, enabling LLMs take on the role of a domain-specific expert who can interpret GNN importance scores within the context of traffic systems. The LLM first evaluates the validity of the user-provided GNN importance scores and then generates context-aware explanations when the inputs are deemed reliable. If the scores are invalid, the LLM flags them and informs the user, instead of producing misleading explanations. This validation step is essential for building trustworthy and responsible LLM-based interfaces in critical domains like traffic management.

II. RELATED WORK AND BACKGROUND

A. Evaluating GNN Explanations in Transportation

Practical GNN application in transportation [1], [12] demands transparency. While XAI methods offer insights via scores (e.g., GNNExplainer [6]), evaluating explanation faithfulness and relevance within the specific ITS context remains a challenge [9]. Standard quantitative metrics often miss semantic nuances crucial for traffic [8]. This necessitates scalable methods that incorporate multimodal context and domain knowledge, motivating our LLM-based approach.

B. LLM Prompting and Reflective Improvement

The strong dependence of LLM outcomes on prompt design is well-established [13]. Structured prompting (e.g., Chain-of-Thought [14]–[16]) can improve LLM reasoning. However, manual prompt design is intractable for complex, dynamic, multimodal tasks like traffic analysis. Static prompts often lack adaptability. Generalisation for automated prompt optimisation is a key challenge [13], [17], though counterfactual methods show potential [18]. Thus, adaptive prompt or agent guidance is needed.

Self-reflection and self-correction mechanisms show promise for enhancing LLM and agent robustness, adaptability, and reliability [19], [20]. Techniques allowing agents to critique outputs and refine prompts (e.g., Reflexion [21], self-critique [22]) improve system capability and dependability [23]. These principles underpin our framework’s methodology.

C. LLM Agents, Multi-Agent Systems, and Trustworthiness

LLM capabilities drive the development of *LLM-based agents* for complex tasks [11]. Collaborative *Multi-Agent Systems (MAS)* organise these agents, enabling sophisticated problem-solving via distributed roles [19], [24]–[27]. However, ensuring system *trustworthiness* is critical, especially in transportation [23]. Concerns about LLM *hallucination* (plausible-but-incorrect outputs) are significant [28]. Using LLMs as evaluators, or *LLM-as-a-judge* [29], is emerging but requires careful design, particularly when input validity is uncertain. Reflection aids reliability [19], [21], [22]. However, evaluating potentially unreliable inputs (like GNN scores) remains challenging because using invalid inputs undermines trustworthiness. This motivates our framework’s **Phase 1 pre-validation step** to establish score reliability before explanation analysis, bolstering overall integrity.

D. LLM Applications in Intelligent Transportation Systems

LLMs are finding diverse ITS roles, including traffic forecasting [30], [31], autonomous driving [32], and controlling simulations [33], [34]. These applications often leverage multimodal capabilities [35], [36]. Our research uses *LLM agents* distinctly: to *interpret and evaluate explanations* from other models (Traffic GNNs). The goal is leveraging their reasoning within our framework to enhance GNN predictor transparency and trustworthiness, not replace them.

III. METHODOLOGY: REFLECTIVE PROMPT OPTIMISATION FRAMEWORK

A. Problem Statement

Traffic Prediction Model: Let the traffic network be represented as a directed graph $G = (\mathcal{N}, \mathcal{A})$, where $\mathcal{N}$ is the set of nodes (e.g., road segments or sensors) and $\mathcal{A}$ is the set of edges (connections between nodes). Each node $i \in \mathcal{N}$ is associated with traffic features over time, such as speed, volume, and occupancy. We define the model input at time t as $\mathbf{X}_t \in \mathbb{R}^{T \times |\mathcal{N}| \times F}$, where T is the historical window size (number of past time steps), $|\mathcal{N}|$ is the number of nodes, and F is the number of features per node (e.g., $F = 3$ for speed, volume, and occupancy). Given the input $\mathbf{X}_t = [\mathbf{x}_{t-T+1}, \dots, \mathbf{x}_t]$, with each $\mathbf{x}_\tau \in \mathbb{R}^{|\mathcal{N}| \times F}$ representing the traffic state at time τ , the model’s goal is to predict a specific traffic feature (e.g., speed) at all nodes for the next time step $t + 1$:

$$\hat{\mathbf{y}}_{t+1} = \mathcal{M}(G, \mathbf{X}_t)$$

where $\mathcal{M}$ is a learned GNN model, and $\hat{\mathbf{y}}_{t+1} \in \mathbb{R}^{|\mathcal{N}|}$ denotes the predicted speeds for all nodes in the network at time $t + 1$.

GNN Explainability (Node-level Importance Scores): To interpret the prediction $\hat{\mathbf{y}}_{t+1}$ made by the model $\mathcal{M}$, post-hoc explainability methods such as GNNExplainer can be applied. These methods seek to learn a local subgraph $G_i = (\mathcal{N}_i, \mathcal{A}_i)$ centred on node i , capturing the most relevant nodes and edges, and an edge importance mask $Mask^{\text{edge}} \in [0, 1]^{|\mathcal{A}_i|}$ indicating the contribution of each edge in $\mathcal{A}_i$. To formalise the influence of neighbouring nodes on the prediction for a target node $i \in \mathcal{N}$, we define an importance score vector $S_i \in \mathbb{R}^{|\mathcal{N}_i|}$, where $\mathcal{N}_i \subseteq \mathcal{N}$ denotes the set of nodes in the local subgraph relevant to node i (typically including i and its k -hop neighbours). Each entry $S_i[j]$ represents the aggregated importance of neighbouring node $j \in \mathcal{N}_i$ in contributing to the prediction for target node i :

$$S_i[j] = \sum_{(j,k) \in \mathcal{A}_i, k=i \text{ or } j=i} Mask_{(j,k)}^{\text{edge}}$$

This importance score vector S_i allows us to identify and rank the most influential neighbouring nodes in the model’s prediction for node i .

LLM-Based Explanation of GNN Model Predictions: Once the importance score vector S_i is obtained for a specific prediction instance for target node i at time t , the next step is to generate a natural language explanation that contextualises the model’s output for that target node $\hat{y}_{t+1}[i]$ (e.g., the predicted speed on a target road segment). Fig. 1 illustrates the concept of using an LLM to interpret GNN traffic predictions using importance scores and multimodal traffic context. First, we construct a prompt for a Large Language Model (LLM) that includes the following multimodal inputs: (1) the importance score vector S_i , indicating the influence of neighbouring nodes, (2) the original input sequence $\mathbf{X}_t$, representing traffic features over the past T time steps, (3) the model’s predicted output $\hat{y}_{t+1}[i]$, and (4) additional context including an image

or map snippet capturing the geographic layout of the local neighbourhood $\mathcal{N}_i$.

The LLM is tasked with explaining why the model predicted, for example, a low speed at node i at time $t + 1$, by reasoning over the influence patterns in S_i . This involves identifying which upstream or adjacent road segments contributed most to the congestion, and articulating how their spatial positioning and traffic conditions affect the target node. For instance, the LLM might infer that “*the predicted slowdown at segment i is likely due to congestion on an upstream segment j_1 , which holds the highest influence score and lies directly on the main inflow path to i . Additionally, segment j_2 , the second most influential neighbour, is located diagonally upstream and is also experiencing high volume, likely compounding the congestion.*” The goal is for the LLM to synthesise these numeric and visual cues into a coherent narrative that reflects the model’s logic, making the prediction more interpretable and actionable for traffic professionals.

B. Proposed Framework Overview

Given the scenario described above, a critical challenge is ensuring that the LLM does not blindly attempt to explain any input it receives, especially when the provided importance scores S_i are invalid or nonsensical. Without a validation mechanism, the LLM may generate fluent but misleading explanations—an issue commonly referred to as *hallucination*. To address this, we propose a **multi-agent reflective prompt optimisation framework** consisting of two sequential phases. In **Phase 1**, we focus on developing a prompt that guides the LLM to critically evaluate the *validity* of an importance score vector before attempting any explanation. This serves as a pre-explanation phase aimed at refining the prompt to best elicit the LLM’s capacity for judgement and discernment in relation to input quality. In **Phase 2**, we perform the complete explanation workflow illustrated in Figure 1, where the LLM is tasked with generating a context-aware narrative explanation based on a GNN model prediction and its associated importance scores. Importantly, Phase 2 incorporates the optimised prompt from Phase 1 as a built-in validation step: the LLM is first instructed to assess whether the input scores are trustworthy. If deemed valid, it proceeds to generate the explanation. If not, it instead warns the user, indicating that the explanation may not be meaningful given the questionable input. This two-phase structure ensures that the LLM’s narrative outputs are grounded in valid reasoning and that the system can proactively flag unreliable model attributions, thereby enhancing trust and interpretability.

Our framework is built around two distinct features and design principles:

(1) Integrated Prompt Optimisation via Multi-Agent Reflection. Unlike conventional uses of LLMs where the prompt is fixed or manually crafted, our framework explicitly integrates automated prompt optimisation into the LLM interaction loop. This is motivated by the observation that designing effective prompts is non-trivial—especially for complex, context-sensitive tasks like traffic explanation—and may

require adaptation to the specifics of each prediction instance. In both phases of our framework, we employ multiple LLM-based agents that engage in an iterative process: one agent generates a response, another reflects on its quality, and a third synthesises improvements to refine the prompt for the next iteration. This reflective loop enables dynamic prompt tuning tailored to the task and input context.

(2) A Dedicated Phase for Validity Assessment Prompt Optimisation. Rather than optimising a single end-to-end explanation prompt during final narrative generation, we introduce a separate phase (Phase 1) dedicated to learning how to assess the validity of GNN importance scores. This separation is critical because it allows us to formulate the validity assessment as a *supervised* task using specially constructed training instances, where valid and invalid score vectors are labelled. This enables quantitative evaluation of the LLM’s ability to distinguish between plausible and implausible importance scores, thereby strengthening the reliability of its judgement during the actual explanation phase (Phase 2). This separation also allows Phase 2 to focus exclusively on optimising the *faithfulness* of the explanation itself, building on a score validation prompt that has been refined in Phase 1 to reliably elicit the LLM’s ability to assess the validity of GNN importance scores.

C. Two-Phase Reflective Prompt Optimisation

We now provide a detailed description of how our framework uses multi-agent LLMs to optimise prompts for two distinct goals. In **Phase 1**, the objective is to refine a prompt that enables the LLM to evaluate the *validity* of GNN-derived importance scores in a traffic context. In **Phase 2**, we build on this foundation to optimise a prompt for generating *faithful* natural language explanations, ensuring they are grounded in validated input scores. Figure 2 illustrate the overall process of this two-phase prompt optimisation.

Phase 1: Prompt Optimisation for Input Validity Assessment: To enable supervised evaluation of the LLM’s ability to judge the quality of importance scores, we construct a synthetic dataset of training instances. For each GNN-generated importance score vector S_i corresponding to a specific traffic prediction instance for target node i at time t , we generate a set of K ‘shuffled’ variants that simulate invalid importance scores. These are denoted as $\tilde{S}_i^{(1)}, \tilde{S}_i^{(2)}, \dots, \tilde{S}_i^{(K)}$, where each $\tilde{S}_i^{(k)}$ retains the same values as S_i but with the element order (ranking) permuted, representing semantically misleading importance score vectors. For example, a shuffled variant might assign high importance to a distant or contextually irrelevant road segment, while ranking a directly connected and highly influential segment unusually low. The original S_i is labelled as **VALID** and each shuffled variant $\tilde{S}_i^{(k)}$ is labelled as **INVALID**, forming a evaluation dataset $\mathcal{D}_i$ for classification:

$$\mathcal{D}_i = \{(S_i, \text{VALID})\} \cup \{(\tilde{S}_i^{(k)}, \text{INVALID}) \mid k = 1, \dots, K\}$$

The LLM is then prompted to classify each score vector as either valid or invalid, based on whether the ranking appears

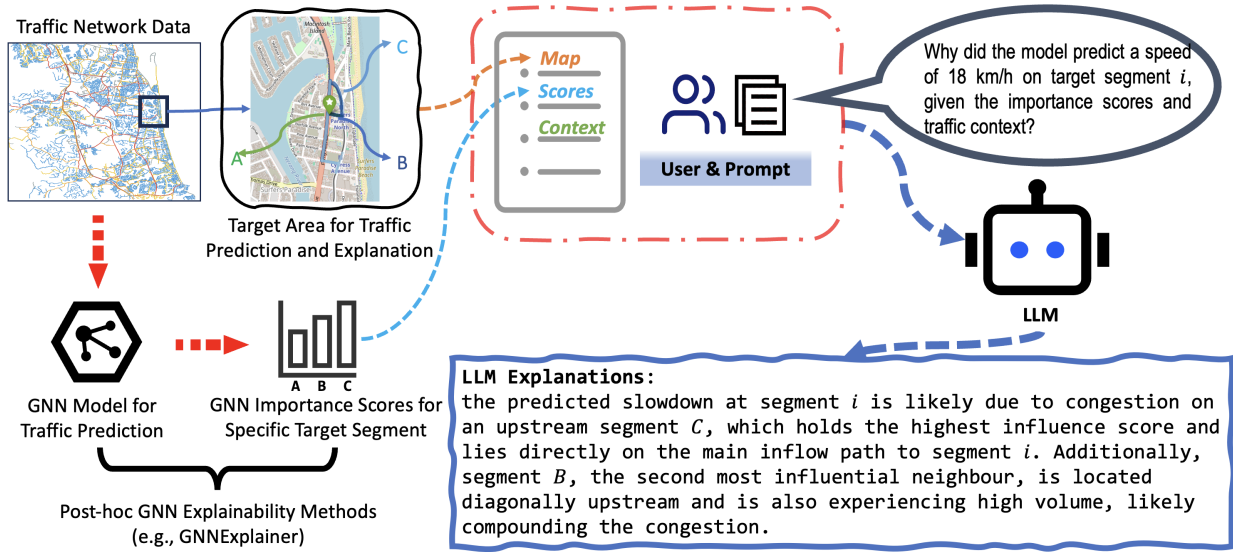


Fig. 1: Conceptual Overview Illustrating the Use of a Large Language Model (LLM) to Generate Narrative Explanations ($\mathcal{N}$) from Graph Neural Network (GNN) importance scores ($\mathcal{S}$)

semantically reasonable within the given traffic context. The prompt includes one score vector $S \in \mathcal{D}_i$, traffic prediction input $\mathbf{X}_t$, prediction output $\hat{y}_{t+1}[i]$, a map image depicting the geographical area covering the target node i 's neighbourhood, and other contextual information such as road segment characteristics, time of day, weather conditions, and known traffic incidents. This classification task is used to evaluate the prompt's effectiveness in eliciting reliable assessments from the LLM. To improve this prompt systematically, we employ a **multi-agent reflective prompt optimisation system** consisting of three collaborative agents:

- **Analysis Agent**, A_{analysis}^1 : Performs classification of each score vector using the current prompt, labelling it as **VALID** or **INVALID**.
- **Reflection Agent**, $A_{\text{reflection}}^1$: Reviews the classification results against the ground truth labels, identifies patterns in the errors, and produces recommendations for improving the prompt.
- **Synthesis Agent**, $A_{\text{synthesis}}^1$: Incorporates the feedback from the reflection agent to revise and generate an improved version of the prompt.

This reflective loop is repeated for M iterations. With each cycle, the prompt becomes better aligned with the LLM's internal reasoning capabilities, improving its ability to critically evaluate the validity of importance scores in diverse traffic scenarios. The output of Phase 1 is a final, optimised prompt that can reliably guide any LLM to perform score validation—serving as a foundation for the explanation process in Phase 2.

Phase 2: Prompt Optimisation for Faithful Explanation Generation

In Phase 2, we focus on generating faithful natural language explanations for GNN predictions using importance scores

from post-hoc GNN explainability methods (e.g., GNNExplainer) and additional contextual traffic information. Unlike Phase 1, which operates in a supervised training setup for prompt refinement, Phase 2 represents the real-world application scenario. Here, the LLM is tasked with explaining a prediction for a new target node i and time step t based on actual model outputs. Each explanation instance consists of the following components: the input and output of a GNN traffic prediction for a target node i , $(\mathbf{X}_t, \hat{y}_{t+1}[i])$, the associated node-level importance score vector S_i , A map image depicting the geographic region covering the neighbourhood of node i , and additional contextual information such as road segment characteristics, time of day, weather, and known traffic incidents.

The explanation prompt used in this phase incorporates the validated prompt structure optimised in Phase 1. Specifically, we embed the Phase 1 prompt components responsible for guiding the LLM to assess input score validity. We first ask the LLM to determine whether the given importance scores appear valid in light of the contextual input, then proceed accordingly. If the scores are deemed valid, the LLM is instructed to generate a faithful explanation grounded in the provided multimodal context. If the scores appear inconsistent or counterintuitive (e.g., highlighting spatially irrelevant nodes or ignoring obvious upstream causes), the LLM is expected to flag the issue and provide reasoning for its concerns, advising further review.

As in Phase 1, we employ a **multi-agent reflective prompt optimisation system** to iteratively refine the Phase 2 prompt to improve the LLM explanations.

The overall Phase 2 prompt is structured in two parts. The first part incorporates the optimised validity-checking instruction developed in Phase 1. This portion of the prompt is *frozen*—that is, it remains unchanged throughout Phase 2 and

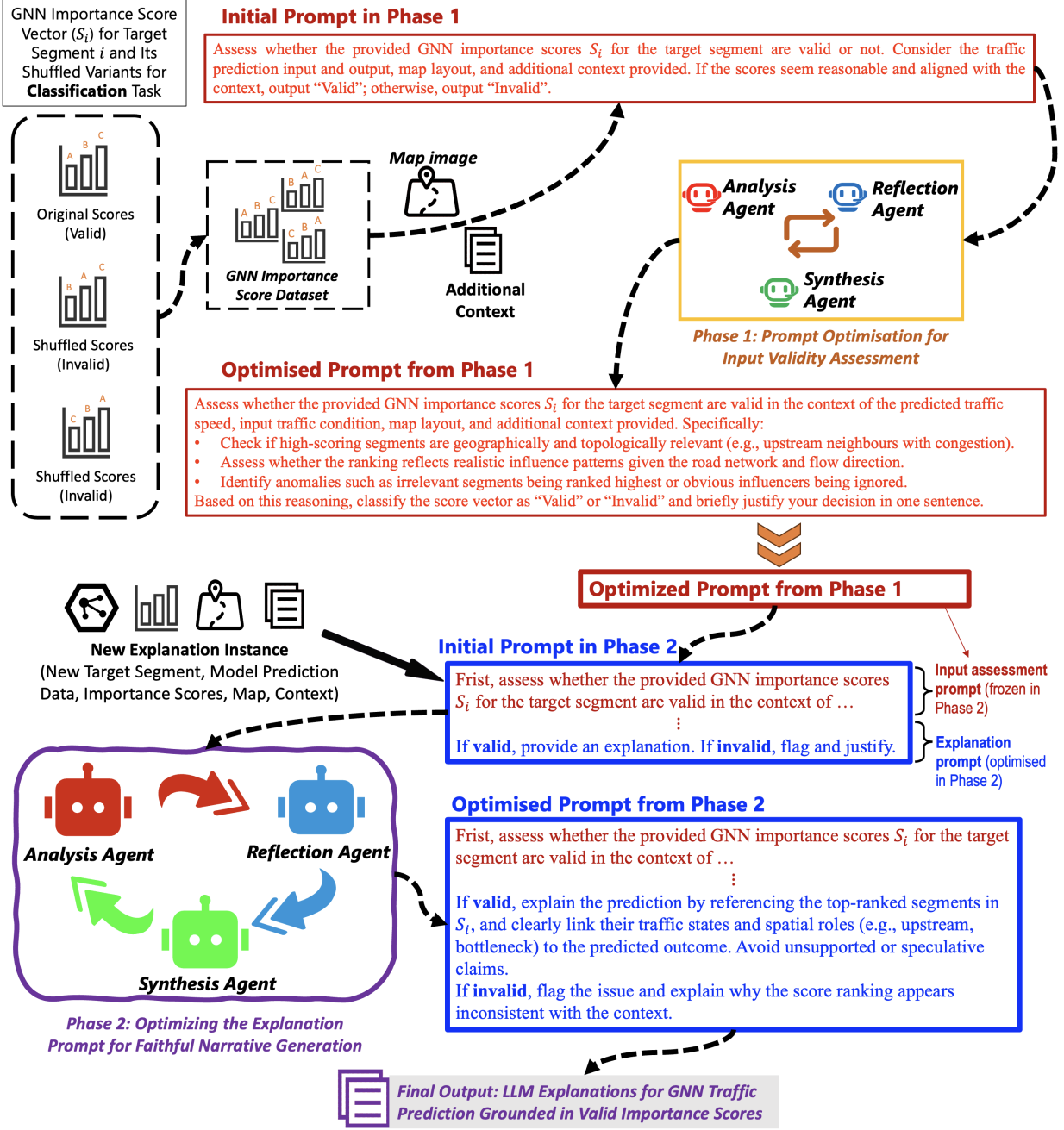


Fig. 2: Overview of Two-Phase Reflective Prompt Optimisation Framework

is reused as-is to ensure consistent and reliable input validation across instances. The second part of the prompt, which is responsible for generating the natural language explanation based on the validated input, focuses on optimisation in Phase 2. Only this explanation portion of the prompt undergoes iterative refinement during this phase, guided by the following three collaborative agents:

- **Analysis Agent**, A_{analysis}^2 : Generates a narrative explanation for each instance using the current prompt.
- **Reflection Agent**, $A_{\text{reflection}}^2$: Reviews the generated ex-

planation, assesses its *faithfulness* with respect to the validated scores and context, and produces a set of recommendations for prompt improvement.

- **Synthesis Agent**, $A_{\text{synthesis}}^2$: Incorporates these recommendations to revise and enhance the prompt for the next iteration.

This reflective optimisation loop is repeated for M' iterations, with the goal of improving the *faithfulness* of the generated explanations across instances. Faithfulness in this context is evaluated based on several key dimensions:

- *Score Grounding*: Whether the explanation explicitly references and aligns with the top-ranked influential nodes in S_i .
- *Contextual Coherence*: Whether the explanation logically integrates geographic, temporal, and environmental context.
- *Causal Plausibility*: Whether the explanation provides a reasonable causal narrative consistent with the traffic network structure and the prediction model’s behaviour.

Through this process, the explanation prompt is iteratively refined throughout this phase, serving as a tool to elicit high-quality responses. Unlike Phase 1—where the final output is an optimised prompt—the final product of Phase 2 is the explanation narrative itself, which is expected to be well-grounded in the validated importance scores and consistent with the traffic context. These narratives are the deliverables intended for use by domain experts and traffic operators.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

This study utilises a Graph Neural Network (GNN) trained on real-world spatio-temporal traffic data from Queensland, Australia. The GNN architecture employs Graph Attention Networks (GATs) [37] with multiple layers and Jumping Knowledge connections (JK) [38] to capture complex dependencies. The GNN predicts road segment traffic speeds using features like traffic volume, occupancy, and segment length. Explanation scores (S) from applying *GNNExplainer* to this pre-trained GNN constitute the primary input instances evaluated by our framework.

Specifically, we evaluated the generalisation capability of the optimised Phase 1 prompt using a dedicated test set ($\mathcal{D}_{test}$) containing $N=200$ distinct traffic explanation instances; **we excluded this set from prompt optimisation**. For the subsequent evaluations focusing on Phase 2’s conditional explanation optimisation, we first applied the finalised Phase 1 prompt to the primary test set to identify instances whose scores were classified as ‘valid.’ We randomly selected $n=500$ for the qualitative analysis, expert validation, and ablation study from this pool of programmatically validated instances. The following subsections detail the performance of our proposed reflective framework under these conditions.

B. Phase 1: Demonstrating Score Validation Prompt Optimisation and Generalisation

Demonstration Goal & Procedure: To demonstrate the effectiveness of Phase 1 optimisation, we prepared a training set $\mathcal{D}_{train} = \bigcup_{i=1}^{1000} \mathcal{D}_i$ containing important score vectors for 1000 different traffic prediction instances. We also prepared a test set $\mathcal{D}_{test} = \bigcup_{i=1}^{500} \mathcal{D}_i$ to test the LLM’s ability to assess the validity of importance scores on 500 new prediction instances that were not used for prompt optimisation. We optimised the initial prompt over $M = 10$ iterations using all the importance score vectors in the training set $\mathcal{D}_{train}$. At each iteration m , we evaluated the prompt’s classification accuracy on the test set ($\mathcal{D}_{test}$) to observe the impact of reflection.

Figure 3 illustrates the observed classification accuracy at each optimisation iteration.

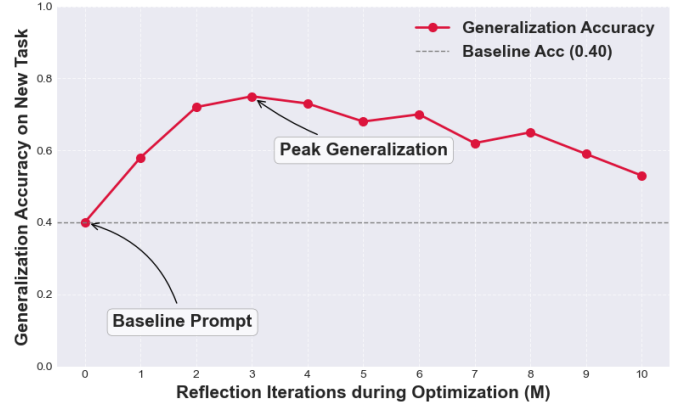


Fig. 3: Demonstration of Phase 1 Reflective Optimisation: Classification accuracy measured on the test set at each optimisation iteration over $M = 10$ steps. The dashed line shows baseline ($M=0$) performance.

Results and Outcome: As shown (Fig. 3), reflection significantly improved the initial prompt from baseline (≈ 0.40) to a peak (≈ 0.75 at $M=3$), demonstrating the effectiveness of the optimisation process in improving the LLM’s ability to assess the validity of GNN importance scores correctly. Qualitative analysis suggested that reflection improved multimodal context integration. We selected the prompt that resulted in the highest accuracy as the final output of Phase 1, using it as the input validator for Phase 2.

C. Phase 2: Conditional Explanation Prompt Optimisation and Faithfulness

The evaluation for Phase 2 aims to show **how** the reflective loop iteratively refines the explanation prompt to increase narrative faithfulness. We focus on assessing the faithfulness of LLM explanations generated under the validated GNN importance scores. Figure 4 exemplifies one reflection cycle (m) within the Phase 2 loop, illustrating how this process enhances faithfulness (the objectives conceptually described in Section III-C). It shows $A_{reflection}^2$ detecting weaknesses, after which $A_{synthesis}^2$ updates the explanation prompt for the next iteration.

Our qualitative analysis of agent logs and outputs across multiple reflection iterations (M') indicated clear improvements. The refined prompts led to narratives that were qualitatively assessed as exhibiting enhanced structure, coherence, and faithfulness. These narratives more effectively grounded explanations in multimodal evidence (e.g., explicitly linking GNN scores to specific map features and contextual traffic information) and addressed shortcomings identified during reflection. Furthermore, expert inspection of 500 test cases confirmed this qualitative gain: faithfulness improved with reflection ($M' > 0$) in 430 instances.

For quantitative analysis, we define an **overlap score** as the proportion of the top-3 most important segments that the

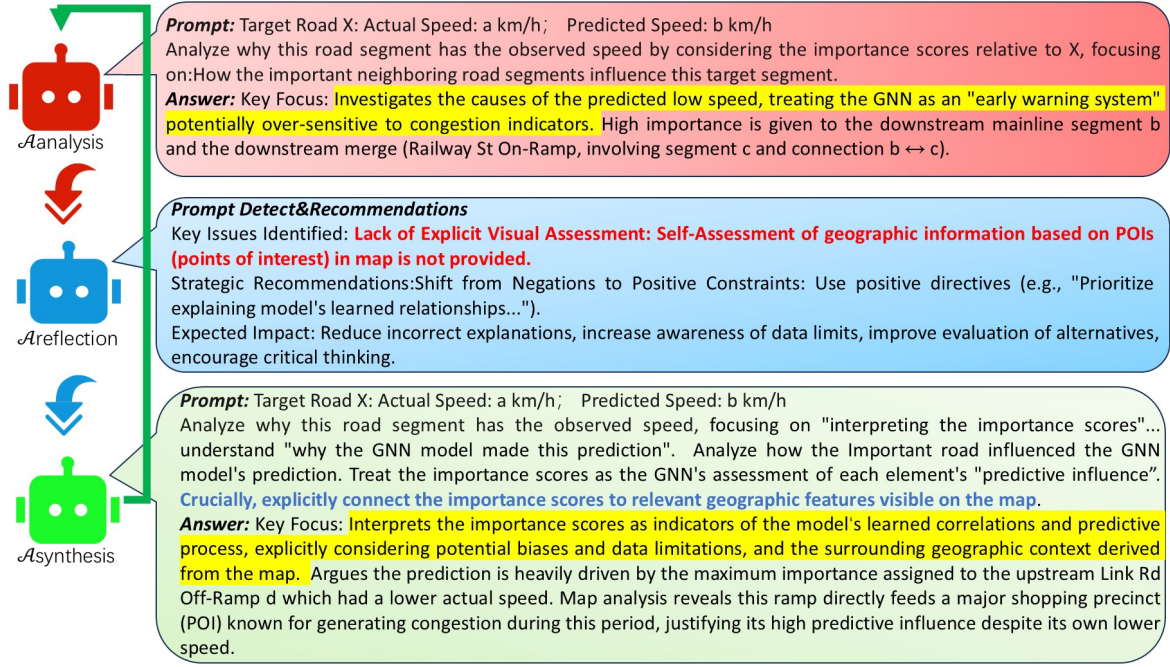


Fig. 4: Example Phase 2 reflection cycle. Top: The initial prompt lacks sufficient grounding in geographic context. Middle: critiques this lack of visual assessment and score linkage. Bottom: produces a refined prompt demanding geographic integration.

narrative mentions explicitly, as proxy metric. The mean overlap score, assessing whether narratives correctly referenced the top-K most important road segments, improved from 0.45 ± 0.20 (baseline) to 0.72 ± 0.15 (optimised). This improved overlap score indicates a significantly better grounding of explanations in specific, relevant road network features crucial for traffic analysis. Additionally, to understand the benefit of having a separate optimisation in Phase 1, we conducted an ablation study that ran Phase 2 without incorporating the result from Phase 1. Removing the Phase 1 validation step degraded narrative faithfulness for potentially invalid scores (dropping from approx. 0.68 to 0.42 on $n=500$ scenarios). This result highlights the risk of generating misleading insights for traffic management without checking score validity. Collectively, the qualitative analysis, ablation study, expert validation, and proxy metric affirm the effectiveness of Phase 2 reflective optimisation. By integrating the assessment of input validity before generating explanations, the framework improves the trustworthiness and interpretability of the final GNN explanations, offering more reliable decision support for transportation professionals than baseline approaches. Developing comprehensive quantitative faithfulness metrics remains a key direction for future work.

V. CONCLUSION AND FUTURE DIRECTIONS

This paper presented a two-phase multi-agent reflective framework to improve the trustworthiness and interpretability of GNN explanations in traffic prediction tasks. Phase 1 focused on optimising a pre-evaluation prompt that enables LLMs to assess the validity of GNN-derived importance scores

through supervised training with synthetic variants. Phase 2 leveraged this validated prompt to guide faithful explanation generation, using a second multi-agent loop to iteratively refine the narrative quality based on contextual grounding, causal plausibility, and alignment with influence scores.

Experiments on real-world traffic data demonstrated that our framework improves both input reliability assessment and explanation faithfulness. Quantitative and qualitative evaluations confirmed that reflective prompt optimisation enhances generalisation, narrative coherence, and domain relevance. By explicitly separating input validation from explanation generation, the framework reduces hallucination risks and provides more dependable decision support for traffic operators.

Future work will explore scalable and automatic faithfulness metrics, test generalisation to other post-hoc GNN explainability metrics and prediction tasks, and improve the robustness of the reflection agents across diverse traffic scenarios.

REFERENCES

- [1] W. Jiang, J. Luo, M. He, and W. Gu, "Graph neural network for traffic forecasting: The research progress," *ISPRS International Journal of Geo-Information*, vol. 12, no. 3, p. 100, 2023.
- [2] W. Jiang and J. Luo, "Graph neural network for traffic forecasting: A survey," *Expert systems with applications*, vol. 207, p. 117921, 2022.
- [3] D. A. Tedjopurnomo, Z. Bao, B. Zheng, F. M. Choudhury, and A. K. Qin, "A survey on modern deep neural network for traffic prediction: Trends, methods and challenges," in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*, pp. 3795–3796, IEEE, 2023.
- [4] E. Dai, T. Zhao, H. Zhu, J. Xu, Z. Guo, H. Liu, J. Tang, and S. Wang, "A comprehensive survey on trustworthy graph neural networks: Privacy, robustness, fairness, and explainability," *Machine Intelligence Research*, vol. 21, no. 6, pp. 1011–1061, 2024.

- [5] H. Zhang, B. Wu, X. Yuan, S. Pan, H. Tong, and J. Pei, "Trustworthy graph neural networks: Aspects, methods, and trends," *Proceedings of the IEEE*, 2024.
- [6] Z. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "Gnnexplainer: Generating explanations for graph neural networks," *Advances in neural information processing systems*, vol. 32, 2019.
- [7] H. Yuan, H. Yu, S. Gui, and S. Ji, "Explainability in graph neural networks: A taxonomic survey," *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 5, pp. 5782–5799, 2022.
- [8] A. Longa, S. Azzolin, G. Santin, G. Cencetti, P. Liò, B. Lepri, and A. Passerini, "Explaining the explainers in graph neural networks: a comparative study," *ACM Computing Surveys*, vol. 57, no. 5, pp. 1–37, 2025.
- [9] L. Faber, A. K. Moghaddam, and R. Wattenhofer, "When comparing to ground truth is wrong: On evaluating gnn explanation methods," in *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining*, pp. 332–341, 2021.
- [10] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong, *et al.*, "A survey of large language models," *arXiv preprint arXiv:2303.18223*, vol. 1, no. 2, 2023.
- [11] Z. Xi, W. Chen, X. Guo, W. He, Y. Ding, B. Hong, M. Zhang, J. Wang, S. Jin, E. Zhou, *et al.*, "The rise and potential of large language model based agents: A survey," *Science China Information Sciences*, vol. 68, no. 2, p. 121101, 2025.
- [12] H. Li, Y. Zhao, Z. Mao, Y. Qin, Z. Xiao, J. Feng, Y. Gu, W. Ju, X. Luo, and M. Zhang, "Graph neural networks in intelligent transportation systems: Advances, applications and trends," *arXiv preprint arXiv:2401.00713*, 2024.
- [13] Y. Liu, J. Singh, G. Liu, A. Payani, and L. Zheng, "Towards hierarchical multi-agent workflows for zero-shot prompt optimization," *arXiv preprint arXiv:2405.20252v2*, 2024.
- [14] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou, *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in neural information processing systems*, vol. 35, pp. 24824–24837, 2022.
- [15] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in neural information processing systems*, vol. 35, pp. 22199–22213, 2022.
- [16] S. Yao, D. Yu, J. Zhao, I. Shafran, T. Griffiths, Y. Cao, and K. Narasimhan, "Tree of thoughts: Deliberate problem solving with large language models," *Advances in neural information processing systems*, vol. 36, pp. 11809–11822, 2023.
- [17] Y. Zhou, A. I. Muresanu, Z. Han, K. Paster, S. Pitis, H. Chan, and J. Ba, "Large language models are human-level prompt engineers," in *The Eleventh International Conference on Learning Representations*, 2022.
- [18] X. Bo, Z. Zhang, Q. Dai, X. Feng, L. Wang, R. Li, X. Chen, and J.-R. Wen, "Reflective multi-agent collaboration based on large language models," *Advances in Neural Information Processing Systems*, vol. 37, pp. 138595–138631, 2024.
- [19] J. Luo, W. Zhang, Y. Yuan, Y. Zhao, J. Yang, Y. Gu, B. Wu, B. Chen, Z. Qiao, Q. Long, *et al.*, "Large language model agent: A survey on methodology, applications and challenges," *arXiv preprint arXiv:2503.21460*, 2025.
- [20] S. Du, J. Zhao, J. Shi, Z. Xie, X. Jiang, Y. Bai, and L. He, "A survey on the optimization of large language model-based agents," *arXiv preprint arXiv:2503.12434*, 2025.
- [21] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," *Advances in Neural Information Processing Systems*, vol. 36, pp. 8634–8652, 2023.
- [22] Z. Gou, Z. Shao, Y. Gong, Y. Shen, Y. Yang, N. Duan, and W. Chen, "Critique: Large language models can self-correct with tool-interactive critiquing," *arXiv preprint arXiv:2305.11738*, 2023.
- [23] M. Yu, F. Meng, X. Zhou, S. Wang, J. Mao, L. Pang, T. Chen, K. Wang, X. Li, Y. Zhang, *et al.*, "A survey on trustworthy llm agents: Threats and countermeasures," *arXiv preprint arXiv:2503.09648*, 2025.
- [24] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, *et al.*, "Autogen: Enabling next-gen llm applications via multi-agent conversation," *arXiv preprint arXiv:2308.08155*, 2023.
- [25] A. Kalyuzhnaya, S. Mityagin, E. Lutsenko, A. Getmanov, Y. Aksenkin, K. Fatkhiev, K. Fedorin, N. O. Nikitin, N. Chichkova, V. Vorona, *et al.*, "Llm agents for smart city management: Enhancing decision support through multi-agent ai systems," *Smart Cities (2624-6511)*, vol. 8, no. 1, 2025.
- [26] T. Nie, J. Sun, and W. Ma, "Exploring the roles of large language models in reshaping transportation systems: A survey, framework, and roadmap," *arXiv preprint arXiv:2503.21411*, 2025.
- [27] K.-T. Tran, D. Dao, M.-D. Nguyen, Q.-V. Pham, B. O'Sullivan, and H. D. Nguyen, "Multi-agent collaboration mechanisms: A survey of llms," *arXiv preprint arXiv:2501.06322*, 2025.
- [28] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, *et al.*, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 1–55, 2025.
- [29] J. Gu, X. Jiang, Z. Shi, H. Tan, X. Zhai, C. Xu, W. Li, Y. Shen, S. Ma, H. Liu, *et al.*, "A survey on llm-as-a-judge," *arXiv preprint arXiv:2411.15594*, 2024.
- [30] X. Guo, Q. Zhang, J. Jiang, M. Peng, M. Zhu, and H. F. Yang, "Towards explainable traffic flow prediction with large language models," *Communications in Transportation Research*, vol. 4, p. 100150, 2024.
- [31] S. Wandelt, C. Zheng, S. Wang, Y. Liu, and X. Sun, "Large language models for intelligent transportation: A review of the state of the art and challenges," *Applied Sciences*, vol. 14, no. 17, p. 7455, 2024.
- [32] Z. Xu, Y. Zhang, E. Xie, Z. Zhao, Y. Guo, K.-Y. K. Wong, Z. Li, and H. Zhao, "Drivegpt4: Interpretable end-to-end autonomous driving via large language model," *IEEE Robotics and Automation Letters*, 2024.
- [33] S. Zhang, D. Fu, W. Liang, Z. Zhang, B. Yu, P. Cai, and B. Yao, "Trafficgpt: Viewing, processing and interacting with traffic foundation models," *Transport Policy*, vol. 150, pp. 95–105, 2024.
- [34] M. Movahedi and J. Choi, "The crossroads of llm and traffic control: A study on large language models in adaptive traffic signal control," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [35] L. Gan, W. Chu, G. Li, X. Tang, and K. Li, "Large models for intelligent transportation systems and autonomous vehicles: A survey," *Advanced Engineering Informatics*, vol. 62, p. 102786, 2024.
- [36] D. Le, A. Yunusoglu, K. Tiwari, M. Isik, and I. Dikmen, "Multimodal llm for intelligent transportation systems," *arXiv preprint arXiv:2412.11683*, 2024.
- [37] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," *arXiv preprint arXiv:1710.10903*, 2017.
- [38] K. Xu, C. Li, Y. Tian, T. Sonobe, K.-i. Kawarabayashi, and S. Jegelka, "Representation learning on graphs with jumping knowledge networks," in *International conference on machine learning*, pp. 5453–5462, PMLR, 2018.